

Gabriel Johnson

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Education

UNIVERSITY OF VERMONT

M.S. Mechanical Engineering

Expected May 2026

- NASA Vermont Space Grant Pre-Doctoral Fellow
- Primary instructor for ME1510 Computational Mech Engr. Lab

B.S. Mechanical Engineering

May 2025

- Research at UVM AIRLab on applying adaptive Kalman filters to autonomous drones (2024-25)
- President/Co-Founder of UVM chapter of the American Society of Mechanical Engineers (2022-25)

UNIVERSIDAD CARLOS III DE MADRID- STUDY ABROAD

August 2023 – June 2024

- Research position focusing on compostable materials in manufacturing under Professor Ángel López.

Work Experience

UNIVERSITY OF VERMONT

Burlington, VT

NASA Predoctoral Fellow

August 2025 - Present

- Developed custom OpenFOAM solvers to model pressure transients and cavitation phenomena in propellant feed systems for NASA Space Grant funded research
- Validated CFD predictions against physical testing at Benchmark Space Systems; applied results to inform pressure transient mitigation and failure mode analysis in propellant feed architectures

BENCHMARK SPACE SYSTEMS

Burlington, VT

Production Engineering Intern

May 2024 - December 2024

- Developed documentation and test fixture designs for propellant tanks and a 1.1 N thruster line.
- Supported propellant feed system integration, hotfire testing, and component qualification; gained hands-on experience with valves, plumbing, and instrumentation in spacecraft fluid systems

LINAMAR NORTH CAROLINA

Asheville, NC

Mechanical Engineering Intern

May 2023 - August 2023

- Created most technical documentation relating to the opening of a friction-stir welding production line.

APT NORTE | CASA INTERAMERICANA

Intag, Ecuador

Water Quality Monitoring Program Lead

June 2025 - August 2025

- Leading a research group in designing a microcontroller-based permanent water quality monitoring station to detect and track water contamination from mining operations.
- Expanded the water quality monitoring program from four to thirty-one waterways

Technical Skills

Software: CFD (ANSYS FLUENT, OpenFOAM), FEA (ANSYS, ABAQUS), CAD (SOLIDWORKS)

Programming: Python, MATLAB, C++, PostgreSQL.

Languages: Spanish, English

Fluid Systems: Propellant feed systems, pressure/flow sizing, transient analysis, water hammer dynamics, cavitation, component integration and testing.